**10.18** Consider the data file *mroz* on working wives. Use the 428 observations on married women who participate in the labor force. In this exercise, we examine the effectiveness of a parent's college education as an instrumental variable.

**a.** Create two new variables. *MOTHERCOLL* is a dummy variable equaling one if *MOTHEREDUC* > 12, zero otherwise. Similarly, *FATHERCOLL* equals one if *FATHEREDUC* > 12 and zero otherwise. What percentage of parents have some college education in this sample?

**b.** Find the correlations between *EDUC*, *MOTHERCOLL*, and *FATHERCOLL*. Are the magnitudes of these correlations important? Can you make a logical argument why *MOTHERCOLL* and *FATHERCOLL* might be better instruments than *MOTHEREDUC* and *FATHEREDUC*?

**c.** Estimate the wage equation in Example 10.5 using *MOTHERCOLL* as the instrumental variable. What is the 95% interval estimate for the coefficient of *EDUC*?

**d.** For the problem in part (c), estimate the first-stage equation. What is the value of the *F*-test statistic for the hypothesis that *MOTHERCOLL* has no effect on *EDUC*? Is *MOTHERCOLL* a strong instrument?

**e.** Estimate the wage equation in Example 10.5 using *MOTHERCOLL* and *FATHERCOLL* as the instrumental variables. What is the 95% interval estimate for the coefficient of *EDUC*? Is it narrower or wider than the one in part (c)?

**f.** For the problem in part (e), estimate the first-stage equation. Test the joint significance of *MOTHERCOLL* and *FATHERCOLL*. Do these instruments seem adequately strong?

**g.** For the IV estimation in part (e), test the validity of the surplus instrument. What do you conclude?

(a)

```
> cat("母親有大學教育者比例：", round(pct_mother, 2), "%\n")
母親有大學教育者比例： 12.15 %
>
> cat("父親有大學教育者比例：", round(pct_father, 2), "%\n")
父親有大學教育者比例： 11.68 %
```

(b)

```
> print(round(corr_matrix, 4))
              educ mothercoll fathercoll
educ        1.0000     0.3595     0.3985
mothercoll  0.3595     1.0000     0.3546
fathercoll  0.3985     0.3546     1.0000
```

| Correlation Range | Interpretation |
|---|---|
| 0.00 - 0.19 | Very Weak Correlation |
| 0.20 - 0.39 | Weak Correlation |
| 0.40 - 0.59 | Moderate Correlation |
| 0.60 - 0.79 | Strong Correlation |
| 0.80 - 1.00 | Very Strong Correlation |

從結果可見，EDUC 與 MOTHERCOLL, FATHERCOLL 的相關係數分別約為 0.36 和 0.40，說明父母大學教育與子女受教程度之間具備適度的相關性。將父母教育簡化為是否具備大學門檻的二元變數，可減少連續年數的極端值影響，也更符合工具變數的外生性要求，因而比原始的教育年數更適合作為工具變數。

According to the conditions for instrumental variables:

1.The instrumental variable should not have a direct effect on the outcome variable (yi) and therefore should not be included as an explanatory variable in the model.

2.It should not be correlated with the error term of the regression (i.e., it should be exogenous).

3.It should be strongly correlated (or at least not weakly correlated) with the endogenous variable (xi).

$MOTHERCOLL$ and $FATHERCOLL$ are more likely to satisfy condition two. This is because they are binary variables indicating whether a parent has some college education, which are less susceptible to endogeneity concerns compared to continuous variables like $MOTHEREDUC$ and $FATHEREDUC$. These binary variables are less likely to be influenced by contemporaneous factors affecting the outcome variable. Therefore, we can conclude that $MOTHERCOLL$ and $FATHERCOLL$ are better instrumental variables than the $MOTHEREDUC$ and $FATHEREDUC$.

(c)

```
Call:
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | mothercoll +
    exper + I(exper^2), data = married_data)

Residuals:
     Min       1Q   Median       3Q      Max
-3.08719 -0.32444  0.04147  0.36634  2.35621

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.1327561  0.4965325  -0.267  0.78932
educ         0.0760180  0.0394077   1.929  0.05440 .
exper        0.0433444  0.0134135   3.231  0.00133 **
I(exper^2)  -0.0008711  0.0004017  -2.169  0.03066 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6703 on 424 degrees of freedom
Multiple R-Squared: 0.147,	Adjusted R-squared: 0.1409
Wald test:   8.2 on 3 and 424 DF,  p-value: 2.569e-05
```

```
> confint(iv_model, level = 0.95)
                    2.5 %        97.5 %
(Intercept) -1.105942034  8.404298e-01
educ        -0.001219763  1.532557e-01
exper        0.017054428  6.963439e-02
I(exper^2)  -0.001658392 -8.385898e-05
```

$\ln(\text{WAGE}) = \beta_1 + \beta_2\text{EXPER} + \beta_3\text{EXPER}^2 + \beta_4\text{EDUC} + \epsilon$

First, we take MOTHERCOLL to substitute EDUC, and get first-stage equation is $\text{EDUC} = \gamma_1 + \theta_1\text{MOTHERCOLL} + v$. Next, we take first-stage equation into orginal model to get second-stage equation is $\ln(\text{WAGE}) = \beta_1 + \beta_2\text{EXPER} + \beta_3\text{EXPER}^2 + \beta_4\text{MOTHERCOLL} + \epsilon^*$

The 95% interval estimate of $\beta_4$ is [-0.0012, 0.1533].

Interpretatinos: In repeated sampling, about 95% interval estimate for the coefficient of EDUC using MOTHERCOLL as the instrumental variable constructed this way will contain the true value of the the parameter $\beta_4$.

(d)

```
Call:
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | mothercoll +
    exper + I(exper^2), data = married_data)

Residuals:
     Min       1Q   Median       3Q      Max 
-3.08719 -0.32444  0.04147  0.36634  2.35621 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)   
(Intercept) -0.1327561  0.4965325  -0.267  0.78932   
educ         0.0760180  0.0394077   1.929  0.05440 . 
exper        0.0433444  0.0134135   3.231  0.00133 **
I(exper^2)  -0.0008711  0.0004017  -2.169  0.03066 * 

Diagnostic tests:
                 df1 df2 statistic  p-value    
Weak instruments   1 424    63.563 1.46e-14 ***
Wu-Hausman         1 423     0.742     0.39    
Sargan             0  NA        NA       NA    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.6703 on 424 degrees of freedom
Multiple R-Squared: 0.147,	Adjusted R-squared: 0.1409 
Wald test:   8.2 on 3 and 424 DF,  p-value: 2.569e-05 
```

The first-stage equation is: $x = \gamma_1 + \theta_1 z + v$

which is $\text{EDUC} = \gamma_1 + \theta_1 \text{MOTHERCOLL} + v$

Estimate the first-stage equation by OLS and obtain the fitted value: $\hat{x} = \hat{\gamma}_1 + \hat{\theta}_1 z$

which is $\widehat{\text{EDUC}} = \hat{\gamma}_1 + \hat{\theta}_1 \text{MOTHERCOLL}$

The F-statistic for the hypothesis that MOTHERCOLL has no effect on EDUC is 63.563. Since 63.563 is greater than the rule of the thumb value of 10, we reject the null hypothesis that the IV which is MOTHERCOLL is weak.

(e)

```
Call:
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | mothercoll +
    fathercoll + exper + I(exper^2), data = married_data)

Residuals:
     Min       1Q   Median       3Q      Max 
-3.07797 -0.32128  0.03418  0.37648  2.36183 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)   
(Intercept) -0.2790819  0.3922213  -0.712  0.47714   
educ         0.0878477  0.0307808   2.854  0.00453 **
exper        0.0426761  0.0132950   3.210  0.00143 **
I(exper^2)  -0.0008486  0.0003976  -2.135  0.03337 * 
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.6679 on 424 degrees of freedom
Multiple R-Squared: 0.153,	Adjusted R-squared: 0.147 
Wald test: 9.724 on 3 and 424 DF,  p-value: 3.224e-06 
```

```
> confint(iv2_model, level = 0.95)
                  2.5 %        97.5 %
(Intercept) -1.04782153  4.896578e-01
educ         0.02751845  1.481769e-01
exper        0.01661839  6.873386e-02
I(exper^2)  -0.00162779 -6.940599e-05
```

$\ln(\text{WAGE}) = \beta_1 + \beta_2 \text{EXPER} + \beta_3 \text{EXPER}^2 + \beta_4 \text{EDUC} + \epsilon$

The 95% interval estimate for the coefficient of EDUC using MOTHERCOLL and FATHERCOLL as the instrumental variables is [0.02752,0.1482].

So we can konw that the 95% interval is slightly narrower than the one in part (c).

(f)

```
Call:
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | mothercoll +
    fathercoll + exper + I(exper^2), data = married_data)

Residuals:
     Min       1Q   Median       3Q      Max
-3.07797 -0.32128  0.03418  0.37648  2.36183

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.2790819  0.3922213  -0.712  0.47714
educ         0.0878477  0.0307808   2.854  0.00453 **
exper        0.0426761  0.0132950   3.210  0.00143 **
I(exper^2)  -0.0008486  0.0003976  -2.135  0.03337 *

Diagnostic tests:
                 df1 df2 statistic p-value
Weak instruments   2 423    56.963  <2e-16 ***
Wu-Hausman         1 423     0.519   0.472
Sargan             1  NA     0.238   0.626
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.6679 on 424 degrees of freedom
Multiple R-Squared: 0.153,	Adjusted R-squared: 0.147
Wald test: 9.724 on 3 and 424 DF,  p-value: 3.224e-06
```

The first-stage equation is $x = \gamma_1 + \theta_1 z_1 + \theta_2 z_2 + v$

which is $\text{EDUC} = \gamma_1 + \theta_1 \text{MOTHERCOLL} + \theta_2 \text{FATHERCOLL} + v$

So we get that $\hat{x} = \hat{\gamma}_1 + \hat{\theta}_1 z_1 + \hat{\theta}_2 z_2$

which is $\widehat{\text{EDUC}} = \hat{\gamma}_1 + \hat{\theta}_1 \text{MOTHERCOLL} + \hat{\theta}_2 \text{FATHERCOLL}$

The F-test statistic of the joint significance of the two IV is 56.963 , is far greater than the rule of thumb value of 10 for a weak IV.

Thus we reject the null hypothesis that the instruments are weak.

(g)

We test the validity of the overidentifying restrictions using the Sargan test. The null hypothesis is that the surplus instruments are valid, i.e., uncorrelated with the error term

The Sargan test statistic is 0.238 with a p-value of 0.626. Since the p-value is greater than 0.05, we fail to reject the null hypothesis. Therefore, the instruments appear to be valid.

**10.20** The CAPM [see Exercises 10.14 and 2.16] says that the risk premium on security $j$ is related to the risk premium on the market portfolio. That is

$$r_j - r_f = \alpha_j + \beta_j(r_m - r_f)$$

where $r_j$ and $r_f$ are the returns to security $j$ and the risk-free rate, respectively, $r_m$ is the return on the market portfolio, and $\beta_j$ is the $j$th security's "*beta*" value. We measure the market portfolio using the Standard & Poor's value weighted index, and the risk-free rate by the 30-day LIBOR monthly rate of return. As noted in Exercise 10.14, if the market return is measured with error, then we face an errors-in-variables, or measurement error, problem.

**a.** Use the observations on Microsoft in the data file *capm5* to estimate the CAPM model using OLS. How would you classify the Microsoft stock over this period? Risky or relatively safe, relative to the market portfolio?

**b.** It has been suggested that it is possible to construct an IV by ranking the values of the explanatory variable and using the rank as the IV, that is, we sort $(r_m - r_f)$ from smallest to largest, and assign the values $RANK = 1, 2, \ldots, 180$. Does this variable potentially satisfy the conditions IV1–IV3? Create *RANK* and obtain the first-stage regression results. Is the coefficient of *RANK* very significant? What is the $R^2$ of the first-stage regression? Can *RANK* be regarded as a strong IV?

**c.** Compute the first-stage residuals, $\hat{v}$, and add them to the CAPM model. Estimate the resulting augmented equation by OLS and test the significance of $\hat{v}$ at the 1% level of significance. Can we conclude that the market return is exogenous?

**d.** Use *RANK* as an IV and estimate the CAPM model by IV/2SLS. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?

**e.** Create a new variable $POS = 1$ if the market return $(r_m - r_f)$ is positive, and zero otherwise. Obtain the first-stage regression results using both *RANK* and *POS* as instrumental variables. Test the joint significance of the IV. Can we conclude that we have adequately strong IV? What is the $R^2$ of the first-stage regression?

**f.** Carry out the Hausman test for endogeneity using the residuals from the first-stage equation in (e). Can we conclude that the market return is exogenous at the 1% level of significance?

**g.** Obtain the IV/2SLS estimates of the CAPM model using *RANK* and *POS* as instrumental variables. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?

**h.** Obtain the IV/2SLS residuals from part (g) and use them (not an automatic command) to carry out a Sargan test for the validity of the surplus IV at the 5% level of significance.

(a)

```
Call:
lm(formula = ex_msft ~ ex_mkt, data = capm5)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.27424 -0.04744 -0.00820  0.03869  0.35801 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.003250   0.006036   0.538    0.591    
ex_mkt      1.201840   0.122152   9.839   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08083 on 178 degrees of freedom
Multiple R-squared:  0.3523,	Adjusted R-squared:  0.3486 
F-statistic:  96.8 on 1 and 178 DF,  p-value: < 2.2e-16
```

微軟的$\hat{\beta}$約等於 1.20 ( t=9.84，p< 2e-16)，顯著大於 1，表示其報酬對市場波動反應更強，屬於比市場更高風險的股票。

(b)

```
Call:
lm(formula = rmrf ~ rank, data = capm5)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.110497 -0.006308  0.001497  0.009433  0.029513 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -7.903e-02  2.195e-03   -36.0   <2e-16 ***
rank         9.067e-04  2.104e-05    43.1   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.01467 on 178 degrees of freedom
Multiple R-squared:  0.9126,    Adjusted R-squared:  0.9121 
F-statistic:  1858 on 1 and 178 DF,  p-value: < 2.2e-16
```

The first-stage regression is mkt_excess = $\pi_0$+ $\pi_1$RANK + $\nu$

The estimated first-stage equation is mkt_excess = -0.0790+ 0.0009067 RANK

The coefficient of RANK is highly statistically significant with a t-statistic of 43.1 and a p-value less than 2e-16. The $R^2$ of the first-stage regression is 0.9126. The F-statistic is 1858, which is far greater than the rule-of-thumb value of 10.

Therefore, RANK can be regarded as a very strong instrumental variable (strong IV).

(c)

```
Call:
lm(formula = msft_excess ~ mkt_excess + vhat, data = capm5)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.27140 -0.04213 -0.00911  0.03423  0.34887 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  0.003018   0.005984   0.504   0.6146    
mkt_excess   1.278318   0.126749  10.085   <2e-16 ***
vhat        -0.874599   0.428626  -2.040   0.0428 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.08012 on 177 degrees of freedom
Multiple R-squared:  0.3672,    Adjusted R-squared:   0.36 
F-statistic: 51.34 on 2 and 177 DF,  p-value: < 2.2e-16
```

We first obtain the first-stage residuals $\hat{\nu}$ from: mkt_excess = $\pi_0$+ $\pi_1$RANK + $\nu$

Then we estimate the augmented CAPM equation:

msft_excess= $\alpha$+ $\beta$ mkt_excess+ $\delta\hat{\nu}$ + e

The estimated coefficient on $\hat{\nu}$ is -0.8746, with a t-statistic of -2.040 and a p-value of 0.0428. Since the p-value is greater than 0.01, we fail to reject the null hypothesis that $\delta = 0$ at the 1% significance level. Therefore, we cannot reject the exogeneity of the market return at the 1% level.

(d)

```
Call:
ivreg(formula = ex_msft ~ ex_mkt | rank, data = capm5)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.271625 -0.049675 -0.009693  0.037683  0.355579 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.003018   0.006044   0.499    0.618    
ex_mkt      1.278318   0.128011   9.986   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08092 on 178 degrees of freedom
Multiple R-Squared: 0.3508,	Adjusted R-squared: 0.3472 
Wald test: 99.72 on 1 and 178 DF,  p-value: < 2.2e-16
```

IV / 2SLS 的 $\beta$≈1.278 ( t=9.99，p < 0.001)，略高於 OLS 的 1.202，符合「市場報酬有測量誤差時，OLS 會向零偏誤」。The IV estimator corrects this bias and produces a larger estimate of beta.

(e)

```
Call:
lm(formula = ex_mkt ~ pos + rank, data = capm5)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.109182 -0.006732  0.002858  0.008936  0.026652 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.0804216  0.0022622  -35.55   <2e-16 ***
pos         -0.0092762  0.0042156   -2.20   0.0291 *  
rank         0.0009819  0.0000400   24.55   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01451 on 177 degrees of freedom
Multiple R-squared:  0.9149,	Adjusted R-squared:  0.9139 
F-statistic: 951.3 on 2 and 177 DF,  p-value: < 2.2e-16
```

We estimate the first-stage regression: mkt_excess = $\pi_0 + \pi_1$RANK + $\pi_2 POS + \nu$

To test the joint significance of the instruments, we test: $H_0 = \pi_1 = \pi_2 = 0$

The F-statistic is 951. 3 with a p-value less than 2.2e-16. Therefore, we reject the null hypothesis that the instruments are jointly insignificant. The $R^2$ of the first-stage regression is 0.9149.

Since the F-statistic is far greater than the rule-of-thumb value of 10, we conclude that RANK and POS are strong instrumental variables.

(f)

We perform a Hausman test by including the first-stage residuals from the regression of mkt_excess on RANK and POS into the CAPM equation:

msft_excess= $\alpha$+ $\beta$ mkt_excess+ $\delta\,\widehat{\nu}$ + e , null hypothesis: $\delta = 0$

```
Call:
lm(formula = ex_msft ~ ex_mkt + v_hat2, data = capm5)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.27132 -0.04261 -0.00812  0.03343  0.34867 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  0.003004   0.005972   0.503   0.6157    
ex_mkt       1.283118   0.126344  10.156   <2e-16 ***
v_hat2      -0.954918   0.433062  -2.205   0.0287 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07996 on 177 degrees of freedom
Multiple R-squared:  0.3696,	Adjusted R-squared:  0.3625 
F-statistic: 51.88 on 2 and 177 DF,  p-value: < 2.2e-16
```

$\widehat{V}$ 的係數約為 -0.955 ( t= -2.205，p= 0.0287)，在 1%水準下不顯著，因此我們無法拒絕「市場超額報酬為外生」的假設。

(g)

msft_excess= $\alpha$+ $\beta$ mkt_excess+ $\delta\,\nu$ + $\widehat{u}$

```
Call:
ivreg(formula = ex_msft ~ ex_mkt | pos + rank, data = capm5)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.27168 -0.04960 -0.00983  0.03762  0.35543 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.003004   0.006044   0.497     0.62    
ex_mkt      1.283118   0.127866  10.035   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08093 on 178 degrees of freedom
Multiple R-Squared: 0.3507,	Adjusted R-squared: 0.347 
Wald test: 100.7 on 1 and 178 DF,  p-value: < 2.2e-16
```

此係數大於 OLS (1.202) 且與單一工具的 IV 結果 (1.283)相近，符合「測量誤差會使 OLS 向零偏誤」。

The IV estimator corrects this bias and produces a higher and more consistent estimate of beta.

The result is also consistent with expectations because the similarity between the IV estimates using RANK alone and RANK + POS suggests that RANK contains most of the explanatory power in the first stage.

(h)

$S = n^* R^2 = 0.5600$

Under the null hypothesis that the instruments are valid, the test statistic follows a chi-square distribution with 1 degree of freedom. The 5% critical value is 3.8414. Since the test statistic is smaller than the critical value, we fail to reject the null hypothesis.

Therefore, we conclude that the instrumental variables RANK and POS are valid at the 5% significance level.